III B.TECH – II SEMESTER

Embedded System Design

Professional Elective-II | JNTUA R23 · B.Tech CSE

	EMBEDDED SYSTEM DESIGN Professional Elective-II	L	T	P	C
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Course Objectives:

- To understand the history, classification, and design process of embedded systems.
- To explore the core components of embedded systems, including processors, memory, and I/O components.
- To introduce onboard and external communication interfaces used in embedded systems.
- To explain different firmware design approaches and programming techniques for embedded systems.
- To provide an understanding of real-time operating systems and task management in embedded systems.

Course Outcomes:

After completing the course, the student will be able to,

- Classify embedded systems based on their purpose, generation, and complexity.
- Identify and select appropriate hardware components for an embedded system design.
- Differentiate and implement various communication protocols like I2C, SPI, and CAN.
- Develop firmware using assembly and high-level programming languages.
- Analyze and apply RTOS-based task scheduling and synchronization techniques.

UNIT I Introduction to Embedded Systems

History of embedded systems, Classification of embedded systems based on generation and complexity, Purpose of embedded systems, The embedded system design process-requirements, specification, architecture design, designing hardware and software, components, system integration, Applications of embedded systems, and characteristics of embedded systems.

UNIT II Typical Embedded System

Core of the embedded system-general purpose and domain specific processors, ASICs, PLDs, COTs; Memory-ROM, RAM, memory according to the type of interface, memory shadowing, memory selection for embedded systems, Sensors, actuators, I/O components: seven segment LED, relay, piezo buzzer, push button switch, other sub-systems: reset circuit, brownout protection circuit, oscillator circuit real time clock, watch dog timer.

** UNIT III Communication Interface **

Onboard communication interfaces-I2C, SPI, CAN, parallel interface; External communication interfaces-RS232 and RS485, USB, infrared, Bluetooth, Wi-Fi, ZigBe, GPRS, GSM.

UNIT IV Embedded Firmware Design and Development

Embedded firmware design approaches-super loop based approach, operating system based approach; embedded firmware development languages-assembly language based development, high level language based development.

UNIT V RTOS based Embedded System Design

Operating system basics, types of operating systems, tasks, process and threads, multiprocessing and multitasking, task scheduling: non-pre-emptive and pre-emptive scheduling; task communication-shared memory, message passing, Remote Procedure Call and Sockets, Task Synchronization: Task Communication/ Synchronization Issues, Task Synchronization Techniques

Text books:

- Introduction to Embedded Systems - Shibu KV, Mc Graw Hill Education.
- Computers as Components –Wayne Wolf, Morgan Kaufmann (second edition).

References:

- Embedded System Design -Frank Vahid, Tony Grivargis, john Wiley.
- Embedded Systems- An integrated approach - Lyla b das, Pearson education 2012.
- Embedded Systems – Raj Kamal, TMH